National Testing Agency

Question Paper Name :	B Tech 8 Apr 2025 Shift 2
Subject Name :	B. Tech
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the values of λ for which the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-\lambda}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 and λ_2. Then the radius of the circle passing through the points $(0,0),(\lambda_1,\lambda_2)$ and (λ_2,λ_1) is

Options :

A. 3

B. $\frac{5\sqrt{2}}{3}$

C. $\frac{\sqrt{2}}{3}$

D. 4

Question Number : 2 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)=x-1$ and $g(x)=e^x$ for $x\in\mathbb{R}$. If $\frac{dy}{dx}=\left(e^{-2\sqrt{x}}\,g\big(f(f(x))\big)-\frac{y}{\sqrt{x}}\right)$, $y(0)=0$, then $y(1)$ is

Options :

A. $\frac{1-e^3}{e^4}$

B. $\frac{e-1}{e^4}$

C. $\frac{1-e^2}{e^4}$

D. $\frac{2e-1}{e^3}$

Question Number : 3 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\ \ldots\infty=\frac{\pi^4}{90}$,

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\ \ldots\infty=\alpha$,

$\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\ \ldots\infty=\beta$,

then $\frac{\alpha}{\beta}$ is equal to :

Options :

A. 23

B. 14

C. 18

D. 15

Question Number : 4 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a be the length of a side of a square OABC with O being the origin. Its side OA makes an acute angle α with the positive x-axis and the equations of its diagonals are $(\sqrt{3}\ +1)x+(\sqrt{3}\ -1)y=0$ and $(\sqrt{3}\ -1)x-(\sqrt{3}\ +1)y+8\sqrt{3}\ =0$. Then a^2 is equal to :

Options :

A. 48

B. 16

C. 24

D. 32

Question Number : 5 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A line passing through the point *P(a, 0)* makes an acute angle α with the positive x-axis. Let this line be rotated about the point *P* through an angle $\frac{\alpha}{2}$ in the clockwise direction. If in the new position, the slope of the line is $2-\sqrt{3}$ and its distance from the origin is $\frac{1}{\sqrt{2}}$, then the value of $3a^2\tan^2\alpha-2\sqrt{3}$ is :

Options :

A. 8

B. 4

C. 5

D. 6

Question Number : 6 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let α be a solution of $x^2+x+1=0$, and for some a and b in R, $\begin{bmatrix}4 & a & b\end{bmatrix}\begin{bmatrix}1 & 16 & 13\\ -1 & -1 & 2\\ -2 & -14 & -8\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\end{bmatrix}$. If $\frac{4}{\alpha^4}+\frac{m}{\alpha^a}+\frac{n}{\alpha^b}=3$, then $m+n$ is equal to _____

Options :

A. 11

B. 3

C. 8

D. 7

Question Number : 7 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the ellipse $3x^2+py^2=4$ pass through the centre C of the circle $x^2+y^2-2x-4y-11=0$ of radius r. Let f_1, f_2 be the focal distances of the point C on the ellipse. Then $6f_1f_2-r$ is equal to

Options :

A. 78

B. 68

C. 70

D. 74

Question Number : 8 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum of the squares of the roots of $|x-2|^2+|x-2|-2=0$ and the squares of the roots of $x^2-2|x-3|-5=0$, is

Options :

A. 24

B. 26

C. 36

D. 30

Question Number : 9 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of $\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right)-\cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$ is equal to

Options :

A. $\pi-\frac{3}{2}$

B. $\pi+\frac{5}{2}$

C. $\pi-\frac{5}{4}$

D. $\pi+\frac{3}{2}$

Question Number : 10 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\left\{\theta\in[0,2\pi]:1+10\,\mathrm{Re}\left(\frac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}\right)=0\right\}$. Then $\sum_{\theta\in A}\theta^2$ is equal to

Options :

A. $\frac{21}{4}\pi^2$

B. $6\pi^2$

C. $\frac{27}{4}\pi^2$

D. $8\pi^2$

Question Number : 11 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: $\lim\limits_{x \to 0}\left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5}\right) = \frac{2}{5}$

Statement II: $\lim\limits_{x \to 1}\left(x^{\frac{2}{1-x}}\right) = \frac{1}{e^2}$

In the light of the above statements, choose the **correct** answer from the options given below:

Options :

A. Statement I is false but Statement II is true

B. Both Statement I and Statement II are false

C. Both Statement I and Statement II are true

D. Statement I is true but Statement II is false

Question Number : 12 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The integral $\int\limits_{-1}^{\frac{3}{2}} \left(|\pi^2 x \sin(\pi x)|\right) dx$ is equal to:

Options :

A. $2 + 3\pi$

B. $4 + \pi$

C. $1 + 3\pi$

D. $3 + 2\pi$

Question Number : 13 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}$.

If $\det(\text{adj}(\text{adj}(3A))) = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, then $m + n$ is equal to

Options :

A. 22

B. 20

C. 24

D. 26

Question Number : 14 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let **A** = {0, 1, 2, 3, 4, 5}. Let **R** be a relation on **A** defined by (x, y) ∈ **R** if and only if max{x, y} ∈ {3, 4}. Then among the statements

(S_1): The number of elements in **R** is 18, and

(S_2): The relation **R** is symmetric but neither reflexive nor transitive

Options :

A. both are false

B. only (S_1) is true

C. only (S_2) is true

D. both are true

Question Number : 15 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let *f(x)* be a positive function and $I_1 = \int_{-\frac{1}{2}}^{1} 2x\, f(2x(1-2x))\, dx$ and $I_2 = \int_{-1}^{2} f(x(1-x))\, dx$. Then the value of $\frac{I_2}{I_1}$ is equal to _____

Options :

A. 12

B. 9

C. 6

D. 4

Question Number : 16 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

Options :

A. 230

B. 210

C. 200

D. 220

Question Number : 17 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If A and B are two events such that $P(A) = 0.7$, $P(B) = 0.4$ and $P(A \cap B) = 0.5$, where B denotes the complement of B, then $P\big(B \mid (A \cup B)\big)$ is equal to

Options :

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{1}{6}$

Question Number : 18 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Let $\hat{c}$ be a unit vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ and be perpendicular to $\vec{a}$. Then such a vector $\hat{c}$ is:

Options :

A. $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{k})$

B. $\frac{1}{\sqrt{5}}(\hat{j} - 2\hat{k})$

C. $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

D. $\frac{1}{\sqrt{3}}(-\hat{i} + \hat{j} - \hat{k})$

Question Number : 19 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the function $f(x) = \frac{x}{3} + \frac{3}{x_5} + 3, x \neq 0$ be strictly increasing in $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ and strictly decreasing in $(\alpha_3, \alpha_4) \cup (\alpha_4, \alpha_5)$. Then $\sum_{i=1}^{} \alpha_i^2$ is equal to

Options :

A. 48

B. 40

C. 36

D. 28

Question Number : 20 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of integral terms in the expansion of $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1016}$ is:

Options :

A. 127

B. 128

C. 130

D. 129

Mathematics Section B

Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The product of the last two digits of $(1919)^{1919}$ is

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 22 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the area of the bounded region $\{(x,y): 0 \leq 9x \leq y^2, y \geq 3x-6\}$ be A. Then $6A$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 23 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let r be the radius of the circle, which touches x - axis at point $(a,0), a<0$ and the parabola $y^2=9x$ at the point (4,6). Then r is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 24 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the area of the triangle formed by the lines $x+2=y-1=z, \frac{x-3}{5}=\frac{y}{-1}=\frac{z-1}{1}$ and $\frac{x}{-3}=\frac{y-3}{3}=\frac{z-2}{1}$ be A. Then A^2 is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 25 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the domain of the function $f(x)=\cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha,\beta]$ and the domain of $g(x)=\log_2\left(2-6\log_{27}(2x+5)\right)$ be (γ,δ).

Then $|7(\alpha+\beta)+4(\gamma+\delta)|$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Physics Section A

Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A monoatomic gas having $\gamma=\frac{5}{3}$ is stored in a thermally insulated container and the gas is suddenly compressed to $\left(\frac{1}{8}\right)^{\text{th}}$ of its initial volume. The ratio of final pressure and initial pressure is:

(γ is the ratio of specific heats of the gas at constant pressure and at constant volume)

Options :

A. 16

B. 32

C. 28

D. 40

Question Number : 27 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. An object is placed at 20 cm to the left of this lens system. The distance of the image from the lens in cm is _____.

Options :

A. $\frac{60}{7}$

B. 15

C. 45

D. 30

Question Number : 28 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A concave-convex lens of refractive index 1.5 and the radii of curvature of its surfaces are 30 cm and 20 cm, respectively. The concave surface is upwards and is filled with a liquid of refractive index 1.3. The focal length of the liquid-glass combination will be

Options :

A. $\frac{700}{11}$ cm

B. $\frac{600}{11}$ cm

C. $\frac{800}{11}$ cm

D. $\frac{500}{11}$ cm

Question Number : 29 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A quantity Q is formulated as $X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. X, Y, and Z are independent parameters which have fractional errors of 0.1, 0.2, and 0.5, respectively in measurement. The maximum fractional error of Q is

Options :

A. 0.6

B. 0.8

C. 0.7

D. 0.1

Question Number : 30 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For a nucleus of mass number A and radius R, the mass density of nucleus can be represented as

Options :

A. $A^{\frac{2}{3}}$

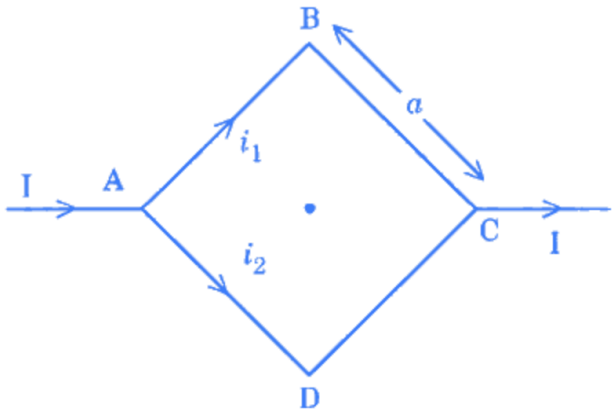
B. Independent of A

C. A^{3}

D. $A^{\frac{1}{3}}$

Question Number : 31 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Figure shows a current carrying square loop ABCD of edge length is 'a' lying in a plane. If the resistance of the ABC part is r and that of ADC part is 2r, then the magnitude of the resultant magnetic field at centre of the square loop is



Options :

A. $\frac{2\mu_0 I}{3\pi a}$

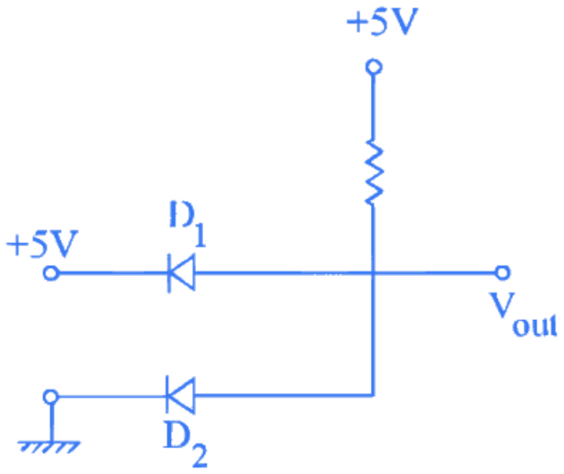
B. $\frac{\sqrt{2}\,\mu_0 I}{3\pi a}$

C. $\frac{3\pi\mu_0 I}{\sqrt{2}\,a}$

D. $\frac{\mu_0 I}{2\pi a}$

Question Number : 32 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The output voltage in the following circuit is (Consider ideal diode case)



Options :

A. 0 V

B. +5 V

C. −5 V

D. 10 V

Question Number : 33 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The amplitude and phase of a wave that is formed by the superposition of two harmonic travelling waves, $y_1(x,t) = 4\sin(kx - \omega t)$ and $y_2(x,t) = 2\sin(kx - \omega t + \frac{2\pi}{3})$, are:

(Take the angular frequency of initial waves same as ω)

Options :

A. $\left[\sqrt{3}, \frac{\pi}{6}\right]$

B. $\left[2\sqrt{3}, \frac{\pi}{6}\right]$

C. $\left[6, \frac{2\pi}{3}\right]$

D. $\left[6, \frac{\pi}{3}\right]$

Question Number : 34 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A 3 m long wire of radius 3 mm shows an extension of 0.1 mm when loaded vertically by a mass of 50 kg in an experiment to determine Young's modulus. The value of Young's modulus of the wire as per this experiment is $P \times 10^{11}\ \text{Nm}^{-2}$, where the value of P is: (Take $g = 3\pi\ \text{m/s}^2$)

Options :

A. 2.5

B. 25

C. 10

D. 5

Question Number : 35 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Water falls from a height of 200 m into a pool. Calculate the rise in temperature of the water assuming no heat dissipation from the water in the pool.

(Take g = 10 m/s^2, specific heat of water = 4200 J/(kg K))

Options :

A. 0.36 K

B. 0.23 K

C. 0.48 K

D. 0.14 K

Question Number : 36 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A rod of linear mass density 'λ' and length 'L' is bent to form a ring of radius 'R'. Moment of inertia of ring about any of its diameter is.

Options :

A. $\frac{\lambda L^3}{8\pi^2}$

B. $\frac{\lambda L^3}{16\pi^2}$

C. $\frac{\lambda L^3}{4\pi^2}$

D. $\frac{\lambda L^3}{12}$

Question Number : 37 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two strings with circular cross section and made of same material, are stretched to have same amount of tension. A transverse wave is then made to pass through both the strings. The velocity of the wave in the first string having the radius of cross section R is v_1, and that in the other string having radius of cross section R/2 is v_2. Then $\frac{v_2}{v_1}$ =

Options :

A. 8

B. 4

C. 2

D. $\sqrt{2}$

Question Number : 38 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two metal spheres of radius **R** and **3R** have same surface charge density **σ**. If they are brought in contact and then separated, the surface charge density on smaller and bigger sphere becomes $\boldsymbol{\sigma_1}$ and $\boldsymbol{\sigma_2}$, respectively. The ratio $\frac{\sigma_1}{\sigma_2}$ is

Options :

A. $\frac{1}{3}$

B. $\frac{1}{9}$

C. 9

D. 3

Question Number : 39 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two balls with same mass and initial velocity, are projected at different angles in such a way that maximum height reached by first ball is 8 times higher than that of the second ball. T_1 and T_2 are the total flying times of first and second ball, respectively, then the ratio of T_1 and T_2 is

Options :

A. 2 : 1

B. $\sqrt{2}$: 1

C. $2\sqrt{2}$: 1

D. 4 : 1

Question Number : 40 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**

Assertion A : Work done in moving a test charge between two points inside a uniformly charged spherical shell is zero, no matter which path is chosen.

Reason R : Electrostatic potential inside a uniformly charged spherical shell is constant and is same as that on the surface of the shell.

In the light of the above statements, choose the correct answer from the options given below.

Options :

A. A is true but R is false

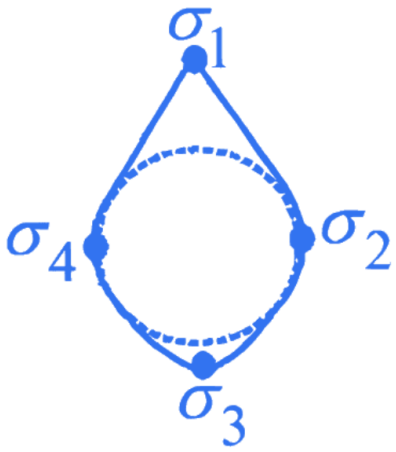
B. Both A and R are true and R is the correct explanation of A

C. Both A and R are true but R is NOT the correct explanation of A

D. A is false but R is true

Question Number : 41 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Electric charge is transferred to an irregular metallic disk as shown in the figure. If σ_1, σ_2, σ_3 and σ_4 are charge densities at given points then, choose the correct answer from the options given below:



A. $\sigma_1 > \sigma_3; \sigma_2 = \sigma_4$

B. $\sigma_1 > \sigma_2; \sigma_3 > \sigma_4$

C. $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$

D. $\sigma_1 < \sigma_3 < \sigma_2 = \sigma_4$

E. $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$

Options :

A. B and C Only

B. A and C Only

C. D and E Only

D. A, B and C Only

Question Number : 42 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An infinitely long wire has uniform linear charge density $\lambda = 2$ nC/m. The net flux through a Gaussian cube of side length $\sqrt{3}$ cm, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be x Nm^2C^{-1}, where x is:

[Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI units]

Options :

A. 6.48π

B. 0.72π

C. 1.44π

D. 2.16π

Question Number : 43 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A body of mass 2 kg moving with velocity of $\vec{v}_{in} = 3\hat{i} + 4\hat{j}$ ms^{-1} enters into a constant force field of 6N directed along positive z-axis. If the body remains in the field for a period of $\frac{5}{3}$ seconds, then velocity of the body when it emerges from force field is.

Options :

A. $3\hat{i} + 4\hat{j} + \sqrt{5}\ \hat{k}$

B. $4\hat{i} + 3\hat{j} + 5\hat{k}$

C. $3\hat{i} + 4\hat{j} - 5\hat{k}$

D. $3\hat{i} + 4\hat{j} + 5\hat{k}$

Question Number : 44 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A block of mass 2 kg is attached to one end of a massless spring whose other end is fixed at a wall. The spring-mass system moves on a frictionless horizontal table. The spring's natural length is 2 m and spring constant is 200 N/m. The block is pushed such that the length of the spring becomes 1 m and then released. At distance x m ($x < 2$) from the wall, the speed of the block will be

Options :

A. $10\left[1 - (2 - x)^2\right]^{\frac{1}{2}}\ m/s$

B. $10\left[1-(2-x)^2\right]^{\frac{3}{2}}\ m/s$

C. $10\left[1-(2-x)^2\right]\ m/s$

D. $10\left[1-(2-x)^2\right]^2\ m/s$

Question Number : 45 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a Young's double slit experiment, the source is white light. One of the slits is covered by red filter and another by a green filter. In this case:

Options :

A. there shall be alternate interference fringes of red and green.

B. there shall be an interference pattern for red distinct from that for green.

C. there shall be an interference pattern, where each fringe's pattern center is green and outer edges is red.

D. there shall be no interference fringes.

Physics Section B

Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A sample of a liquid is kept at 1 atm. It is compressed to 5 atm which leads to a change of volume of 0.8 cm^3. If the bulk modulus of the liquid is 2 GPa, the initial volume of the liquid was _____ litre.

(Take 1 atm = 10^5 Pa)

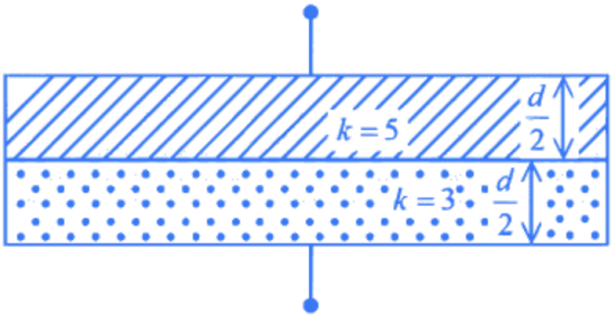
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 47 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A thin solid disk of 1 kg is rotating along its diameter axis at the speed of 1800 rpm . By applying an external torque of 25π Nm for 40 s , the speed increases to 2100 rpm . The diameter of the disk is _____ m.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 48 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript



Space between the plates of a parallel plate capacitor of plate area 4 cm^2 and separation of 1.77 mm, is filled with uniform dielectric materials with dielectric constants (3 and 5) as shown in figure. Another capacitor of capacitance 7.5 pF is connected in parallel with it. The effective capacitance of this combination is _ pF.

(Given $\epsilon_0 = 8.85 \times 10^{-12}$ F/m)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 49 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An electron is released from rest near an infinite non-conducting sheet of uniform charge density '$-\sigma$'. The rate of change of de-Broglie wave length associated with the electron varies inversely as n^{th} power of time. The numerical value of n is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 50 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A cube having a side of 10 cm with unknown mass and 200 gm mass were hung at two ends of a uniform rigid rod of 27 cm long. The rod along with masses was placed on a wedge keeping the distance between wedge point and 200 gm weight as 25 cm. Initially the masses were not at balance. A beaker is placed beneath the unknown mass and water is added slowly to it. At given point the masses were in balance and half volume of the unknown mass was inside the water.

(Take the density of unknown mass is more than that of the water, the mass did not absorb water and water density is 1 gm/cm^3.)

The unknown mass is _____ kg.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText

Chemistry Section A

Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The atomic number of the element from the following with lowest 1st ionisation enthalpy is :

Options :

A. 32

B. 35

C. 19

D. 87

Question Number : 52 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

LIST-I (Complex/ Species)	**LIST-II (Shape & magnetic moment)**
A. $[Ni(CO)_4]$	I. Tetrahedral, 2.8 BM
B. $[Ni(CN)_4]^{2-}$	II. Square planar, 0 BM
C. $[NiCl_4]^{2-}$	III. Tetrahedral, 0 BM
D. $[MnBr_4]^{2-}$	IV. Tetrahedral, 5.9 BM

Choose the **correct** answer from the options given below:

Options :

A. A-I, B-II, C-III, D-IV

B. A-IV, B-I, C-III, D-II

C. A-III, B-II, C-I, D-IV

D. A-III, B-IV, C-II, D-I

Question Number : 53 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the LIST-I with LIST-II

LIST-I	**LIST-II**
A. Carbocation	I. Species that can supply a pair of electrons.
B. C-Free radical	II. Species that can receive a pair of electrons.
C. Nucleophile	III. sp^2 hybridized carbon with empty p-orbital.
D. Electrophile	IV. sp^2/sp^3 hybridized carbon with one unpaired electron.

Choose the **correct** answer from the options given below:

Options :

A. A-IV, B-II, C-III, D-I

B. A-II, B-III, C-I, D-IV

C. A-III, B-IV, C-I, D-II

D. A-III, B-IV, C-II, D-I

Question Number : 54 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one eighth of its initial concentration are t_1 and t_2 (s), respectively. The ratio t_1/t_2 will be:

Options :

A. $\frac{4}{3}$

B. $\frac{3}{2}$

C. $\frac{3}{4}$

D. $\frac{2}{3}$

Question Number : 55 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Correct statements for an element with atomic number 9 are:

A. There can be 5 electrons for which $m_s = +\frac{1}{2}$ and 4 electrons for which $m_s = -\frac{1}{2}$.

B. There is only one electron in p_z orbital.

C. The last electron goes to orbital with $n = 2$ and $l = 1$.

D. The sum of angular nodes of all the atomic orbitals is 1.

Choose the correct answer from the options given below:

Options :

A. A and B Only

B. A, C and D Only

C. A and C Only

D. C and D Only

Question Number : 56 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

The freezing point depression of a 0.1 m aqueous solution of a monobasic weak acid HA is 0.20 °C. The dissociation constant for the acid is

Given: $K_f(H_2O)$ = 1.8 K kg mol^{-1}, molality ≡ molarity

Options :

A. 1.90×10^{-3}

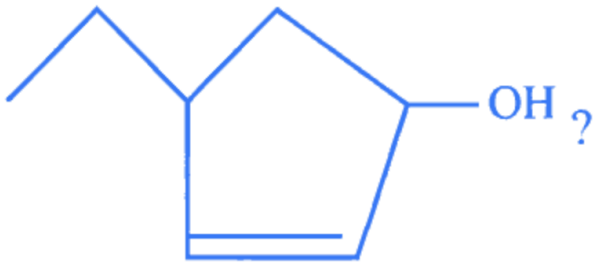
B. 1.38×10^{-3}

C. 1.1×10^{-2}

D. 1.89×10^{-1}

Question Number : 57 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

What is the correct IUPAC name of the compound?



Options :

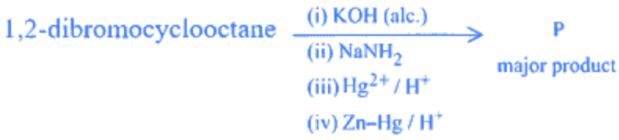
A. 1-Ethylcyclopent-2-en-3-ol

B. 4-Ethylcyclopent-2-en-1-ol

C. 1-Ethyl-3-hydroxycyclopent-2-ene

D. 4-Ethyl-1-hydroxycyclopent-2-ene

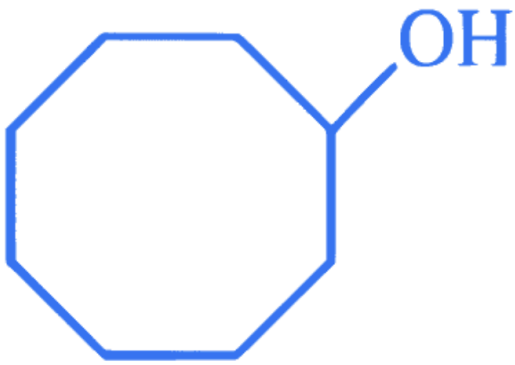
Question Number : 58 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical



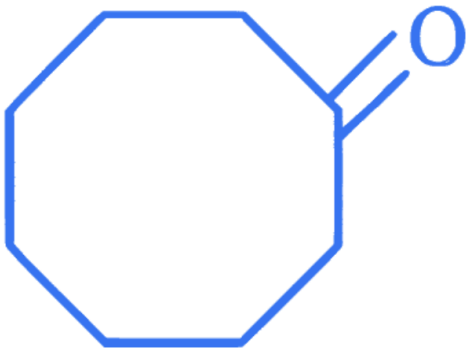
'P' is

Options :

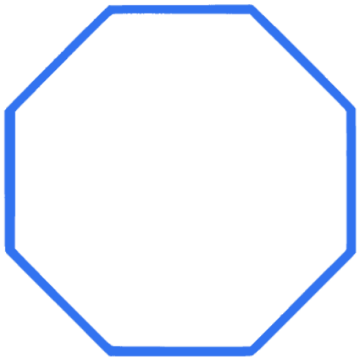
A.



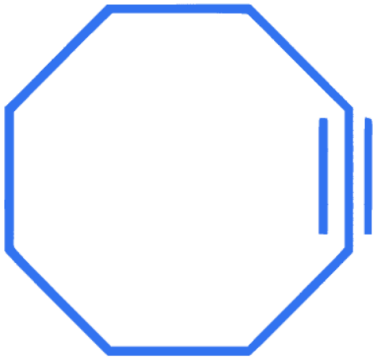
B.



C.



D.



Question Number : 59 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

On combustion 0.210 g of an organic compound containing C, H and O gave 0.127 g H_2O and 0.307 g CO_2.

The percentages of hydrogen and oxygen in the given organic compound respectively are:

Options :

A. 7.55, 43.85

B. 6.72, 53.41

C. 6.72, 39.87

D. 53.41, 39.6

Question Number : 60 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of species from the following that are involved in sp^3d^2 hybridization is :

$[Co(NH_3)_6]^{3+}$, SF_6, $[CrF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$, and $[MnCl_6]^{3-}$

Options :

A. 4

B. 3

C. 5

D. 6

Question Number : 61 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following binary mixture does not show the behaviour of minimum boiling azeotropes?

Options :

A. $CH_3OH + CHCl_3$

B. $C_6H_5OH + C_6H_5NH_2$

C. $H_2O + CH_3COC_2H_5$

D. $CS_2 + CH_3COCH_3$

Question Number : 62 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: H_2Se is more acidic than H_2Te

Statement II: H_2Se has higher bond enthalpy for dissociation than H_2Te

In the light of the above statements, choose the correct answer from the options given below:

Options :

A. Both Statement I and Statement II are true

B. Statement I is true but Statement II is false

C. Both Statement I and Statement II are false

D. Statement I is false but Statement II is true

Question Number : 63 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$$A \xrightarrow[(i)\ H_3O^+]{(i)\ NaH} B \xrightarrow[(ii)\ H_2SO_4,\Delta]{(i)\ EOH} C$$

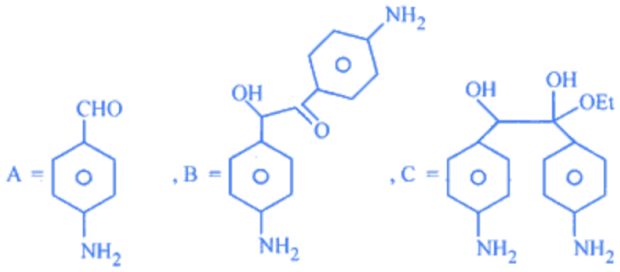
' A ' shows positive Lassaign's test for N and its molar mass is 121 .

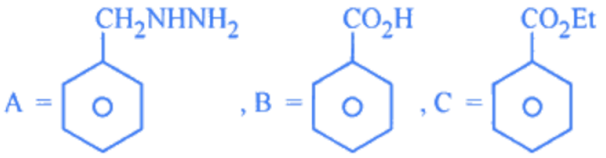
' B ' gives effervescence with aq $NaHCO_3$.

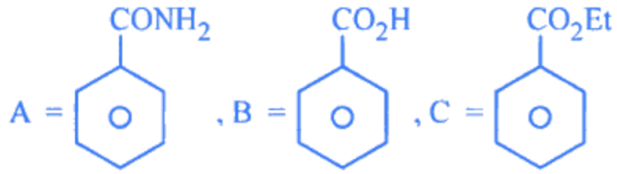
'C' gives fruity smell.

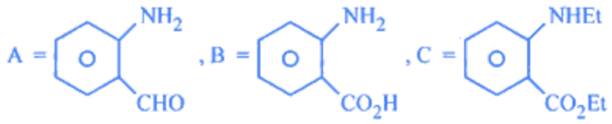
Identify A, B and C from the following.

Options :

A. 

B. 

C. 

D. 

Question Number : 64 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The correct decreasing order of spin only magnetic moment values (BM) of Cu^+, Cu^{2+}, Cr^{2+} and Cr^{3+} ions is :

Options :

A. $Cr^{3+} > Cr^{2+} > Cu^+ > Cu^{2+}$

B. $Cu^+ > Cu^{2+} > Cr^{3+} > Cr^{2+}$

C. $Cr^{2+} > Cr^{3+} > Cu^{2+} > Cu^+$

D. $Cu^{2+} > Cu^{+} > Cr^{2+} > Cr^{3+}$

Question Number : 65 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: A homoleptic octahedral complex, formed using monodentate ligands, will not show stereoisomerism.

Statement II: cis- and trans- platin are heteroleptic complexes of Pd.

In the light of the above statements, choose the **correct** answer from the options given below:

Options :

A. Statement I is true but Statement II is false

B. Both Statement I and Statement II are true

C. Both Statement I and Statement II are false

D. Statement I is false but Statement II is true

Question Number : 66 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

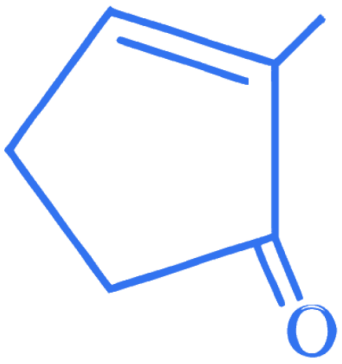
When



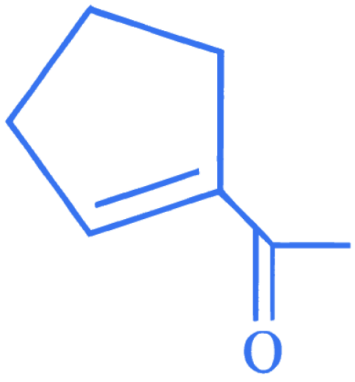
undergoes intramolecular aldol condensation, the major product formed is :

Options :

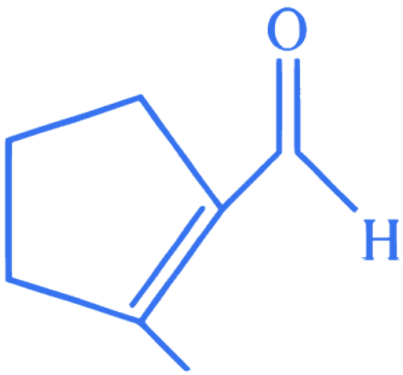
A.



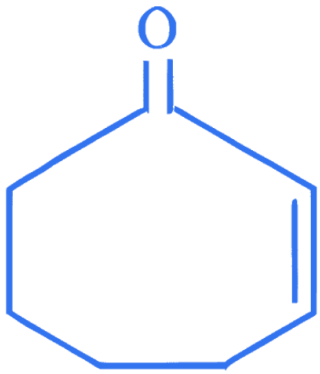
B.



C.

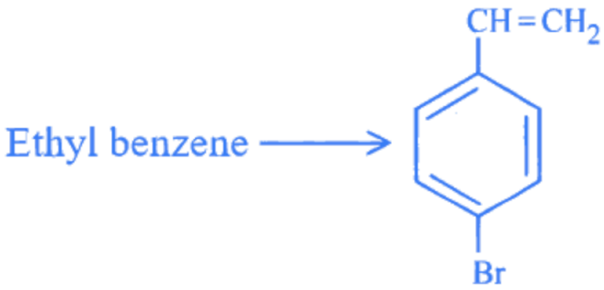


D.



Question Number : 67 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Choose the correct set of reagents for the following conversion.



Options :

A. Cl_2/ anhy · $AlCl_3$; Br_2/Fe; alc. KOH

B. Cl_2/Fe; Br_2/ anhy. $AlCl_3$; aq. KOH

C. $Br_2/Fe; Cl_2, \Delta$; alc. KOH

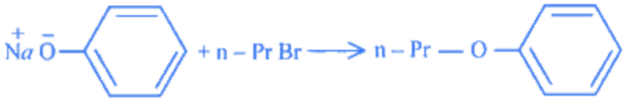
D. Br_2/ anhy. $AlCl_3; Cl_2, \Delta$; aq. KOH

Question Number : 68 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

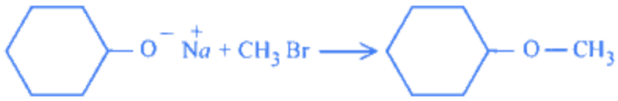
Which one of the following reactions will not lead to the desired ether formation in major proportion?

(iso- Bu ⇒ isobutyl, sec −Bu ⇒ sec-butyl, nPr ⇒ n-propyl,
^{t}Bu ⇒ tert-butyl, Et ⇒ ethyl)

Options :

A. 

B. iso- $BuO\overset{+}{N}a$ + sec – BuBr ⟶ sec- Bu – O – iso -Bu

C. 

D. ^{t}BuO$\overset{+}{N}$a + EtBr ⟶ ^{t}Bu – O – Et

Question Number : 69 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

LIST-I (Reagent)	**LIST-II (Functional Group detected)**
A. Sodium bicarbonate solution	I. double bond/unsaturation
B. Neutral ferric chloride	II. carboxylic acid
C. ceric ammonium nitrate	III. phenolic - OH
D. alkaline $KMnO_4$	IV. alcoholic - OH

Choose the *correct* answer from the options given below:

Options :

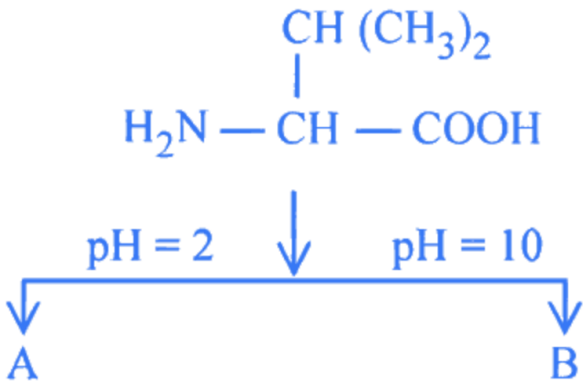
A. A-II, B-III, C-I, D-IV

B. A-II, B-IV, C-III, D-I

C. A-III, B-II, C-IV, D-I

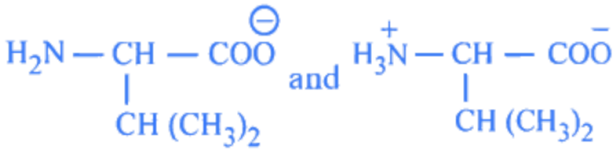
D. A-II, B-III, C-IV, D-I

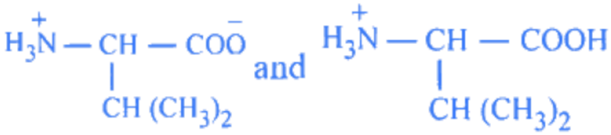
Question Number : 70 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

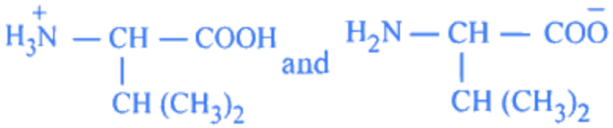


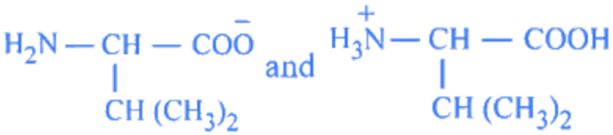
Choose the correct option for structures of *A* and *B*, respectively

Options :

A. 

B. 

C. 

D. 

Chemistry Section B

Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following half cell reaction

$$Cr_2O_7^{2-}(aq) + 6e^- + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

The reaction was conducted with the ratio of

$$\frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}]} = 10^{-6}$$

. The pH value at which the EMF of the half cell will become zero is _____.

(nearest integer value)

[Given: standard half cell reduction potential

$$E^{\circ}_{Cr_2O_7^{2-}, H^+/Cr^{3+}} = 1.33\,V$$

,

$$\frac{2.303RT}{F} = 0.059\,V$$

.]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 72 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The energy of an electron in the first Bohr orbit of the H-atom is -13.6 eV.

The magnitude of energy value of an electron in the first excited state of Be^{3+} is _____ eV (nearest integer value).

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 73 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The equilibrium constant for decomposition of $H_2O(g)$

$$H_2O(g) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g) \quad (\Delta G^{\circ} = 92.34\,kJ\,mol^{-1})$$

is 8.0×10^{-3} at 2300 K and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is _____ $\times 10^{-2}$ (nearest integer value).

[Assume α is negligible with respect to 1]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 74 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

20 mL of sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____ M. (Nearest Integer value)

(Given : Na = 23, I = 127, Ag = 108, N = 14, O = 16 g mol^{-1})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 75 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Resonance in X_2Y can be represented as



The enthalpy of formation of X_2Y $\left(X = X(g) + \frac{1}{2}Y = Y(g) \rightarrow X_2Y(g)\right)$ is 80 kJ mol^{-1}. The magnitude of resonance energy of X_2Y is __ kJ mol^{-1} (nearest integer value).

Given: Bond energies of $X \equiv X$, $X = X$, $Y = Y$ and $X = Y$ are 940, 410, 500, and 602 kJ mol^{-1} respectively. valence X: 3, Y: 2

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText